Forces: what pushes and pulls

LEARNING TARGET · K2 I can name the forces in a scenario: gravity, normal, friction, applied, tension.

A · KEY VOCABULARY

Interaction	Every force has a source (what pushes or pulls) and a receiver.
Contact force	Normal (surface, ⊥), friction (along surface, opposes slip), tension (rope), applied (a push).
Non-contact force	Gravity acts without touching.
Mass vs. weight	Mass is amount of matter (kg). Weight is Earth's gravitational pull (N).

B · KEY EQUATION + UNITS

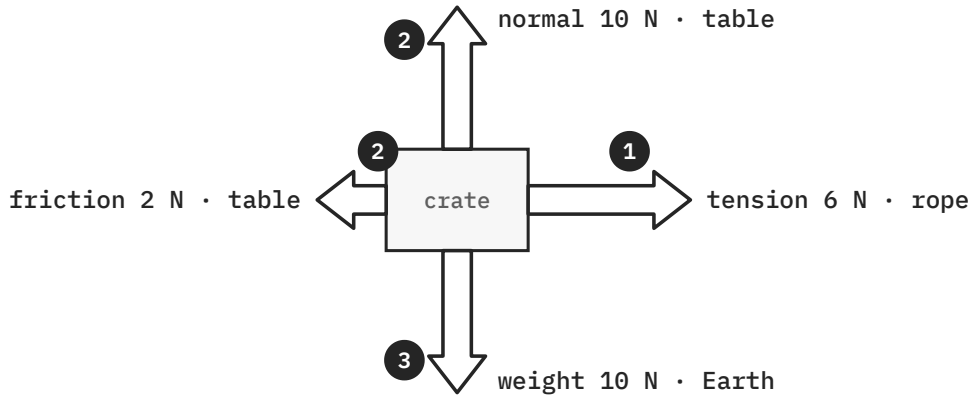
$$F_g = m\ g$$

g = 9.8 N/kg (≈ 10 to estimate)

m mass, kilograms (kg)
W weight, newtons (N)

C · COLOR IT · NAME EVERY SOURCE

The ___ exerts a ___ force on the crate.



☐ **1 · ORANGE** Force from the rope

☐ **2 · ORANGE** Forces from the table

☐ **3 · ORANGE** Force from Earth

D · WORKED EXAMPLE

A 3 kg object rests on a level table. Find its weight and name the force that supports it.

GIVEN	EQUATION	WORK	ANSWER
m = 3 kg g ≈ 10 N/kg	W = m × g	W = 3 × 10 = 30 N down table pushes up equally	**30 N** normal force 30 N up

E · THE 2026-XJ THREAD

DAY 10 OF 22 · NEW TOOL: NAMING FORCES

If XJ's path ever bends or speeds up, something is pulling on it. Naming the source of every force (out there, mostly gravity) is how analysts explain a change.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else